AI-Powered Automated Waste Segregation & E-Waste Recycling System

Proposed IoT System

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1. Introduction

This report documents the complete Internet of Things (IoT) hardware layer of the AI-Powered Automated Waste Segregation and E-Waste Recycling System. The overall system is built from four cooperating components, each contributed by one member of the project team and each responsible for a distinct stage of the waste-handling pipeline: environmental hazard monitoring, plastic-generation sorting, AI-based waste assessment, and predictive economic valuation with strategic disposition.

Although each component runs its own edge controller and sensor set, all four are designed to operate as a single integrated system: every controller reports its readings to a shared FastAPI-based backend, which performs the classification, forecasting, and decision logic, and returns instructions that drive the physical actuators (servos, LEDs, buzzers, displays) back at the edge. This report describes each component's hardware, wiring, and data flow individually, then summarizes how the four are tied together.

2. System Architecture Overview

The system follows a consistent pattern at every stage: sensors capture a physical property of the waste item, an edge microcontroller performs light preprocessing and local alerting, the reading is transmitted over Wi-Fi (or serial, for Component 3) to a backend service, and the backend's decision is sent back to the edge device to drive an actuator, display, or alert. The four components and their primary role in this pipeline are summarized below.

Component	Primary Role	Edge Controller(s)
Component 1	Environmental hazard monitoring (gas, smoke, CO, temperature, humidity)	ESP32 DevKit V1
Component 2	Plastic-generation sorting via moisture and colour sensing	ESP32
Component 3	Weight-triggered image capture and AI waste-type/condition assessment	Arduino UNO + Raspberry Pi + ESP32-CAM
Component 4	Predictive metal valuation and strategic (pyrolysis) disposition	ESP32 DevKit

A detailed system architecture diagram (covering every sensor, actuator, GPIO pin, and inter-component data flow) is provided separately as an editable draw.io / Mermaid diagram accompanying this report.

3. Component 1 — Air Quality & Environmental Hazard Monitoring

This component continuously monitors the ambient air condition around the waste-handling station and raises a local alert when gas or smoke readings become elevated.

3.1 Controller

ESP32 DevKit V1 acts as the central edge-computing controller. It reads all sensors, performs basic preprocessing, determines the local alert status, drives the LCD/LED/buzzer outputs, and transmits measurements over both Wi-Fi (to the backend) and MQTT (publish).

3.2 Sensors and Actuators

Component	Connection	Purpose
DHT22	Digital pin	Measures temperature and relative humidity; used as environmental context since MQ-series responses vary with these conditions.
MQ-2	Analog (ADC)	Combustible gas / smoke response. Recorded as the raw value mq2_raw — not converted to ppm without calibration.
MQ-135	Analog (ADC)	Cross-sensitive air-quality/VOC response (including ammonia). Recorded as mq135_raw.

MQ-7	AO → GPIO 35	CO-related sensing. Requires proper heater operation/cycling for a meaningful CO reading; treated as a raw sensor response until that calibration procedure is implemented.
I2C LCD	SDA → GPIO 21, SCL → GPIO 22	Local display of MQ readings, temperature, humidity, and current alert state.
Green LED	GPIO 14	Normal condition.
Yellow LED	GPIO 27	Caution / elevated sensor response.
Red LED	GPIO 26	High sensor response.
Buzzer	GPIO 13	Local audible warning.

Design note: the raw-ADC alert thresholds used by this component describe relative sensor response only. They must not be reported as WHO exposure limits unless a proper calibrated relationship between raw ADC output and a recognised gas concentration standard is established.

4. Component 2 — SHEF Plastic Sorting Module

This module (Sorting by Humidity, Ejection and colour-based Fading — SHEF) connects four sensing/actuation elements to a single ESP32 to sort plastic items by moisture content and by the degree of thermal yellowing ("generation") of the material.

4.1 Hardware and Wiring

Component	**Connection**	**Purpose**
Capacitive Moisture Sensor	VCC → 3V3, GND → GND, AOUT → D34	Detects Wet or Dry by measuring the capacitance change when it contacts the material's surface.
Servo Motor (sorting actuator)	Power/GND from shared VIN rail; Signal → D13	Executes the final sorting action by rotating to the angle instructed by the backend's process recipe.
HC-SR04 Ultrasonic	Trig → D5, Echo → D18 (shared power rail)	Confirms an item is present before processing begins; measures distance from pulse round-trip time.
TCS34725 Colour Sensor	I2C — SDA → D21, SCL → D22	Measures RGB values of the plastic sample to estimate thermal yellowing, classified into Generation 1, 2, or 3 by firmware threshold logic.

All four components share a common ground back to the ESP32. Because the ESP32's VIN pin has only one physical socket, the servo and the HC-SR04 draw power from a shared breadboard rail fed from VIN.

4.2 Data Flow

In its main loop the ESP32 reads moisture, distance, and colour data, then sends this reading over Wi-Fi as JSON to the FastAPI backend. The backend returns a process recipe, and the servo motor executes the corresponding sorting action.

5. Component 3 — Weight-Triggered Image Capture & AI Waste Assessment

This component captures the physical and visual data needed for AI-based waste assessment before an item is routed to its grade-specific bin.

5.1 Hardware

Device	Role	Interface
Load Cell + HX711	Measures the weight of the waste item	Analog load cell → HX711 amplifier → Arduino UNO
Arduino UNO	Reads the HX711 weight data and forwards it onward	Serial (UART) → Raspberry Pi
Raspberry Pi	Main edge controller; manages the weight-triggered image-capture workflow	Serial in (weight) / Wi-Fi or local link (camera + API)
ESP32-CAM	Captures the waste image on request	Requested by the Raspberry Pi
Waste Assessment API	AI prediction service	Receives image via HTTP; returns waste type, condition, confidence, decision status

5.2 Data Flow

The Arduino UNO reads the load-cell data through the HX711 module and sends the weight readings to the Raspberry Pi over serial communication. The Raspberry Pi monitors this weight stream, and once the measured weight passes the defined threshold, it requests an image from the ESP32-CAM and forwards the captured image to the Waste Assessment API. The API returns the waste type, condition, confidence values, and decision status, which the Raspberry Pi stores for monitoring and further analysis.

5.3 Physical Sorting Layout

The station is arranged around two conveyor belts. Conveyor Belt 1 carries the incoming waste item to the weight-sensor station (Sensor Module 1 — the load cell); the weight is measured there and the waste image is captured in the same step. Once the AI assessment returns the item's grade, Conveyor Belt 2 (guided by Sensor Module 2) diverts the item into the collection bin that corresponds to its predicted grade (labelled A, B and C in the design sketch).

6. Component 4 — Predictive Economic Valuation & Strategic Disposition Terminal

This component decides, for each ingested item, whether recovered base metal should be sold immediately or held and compacted, and — for non-recyclable polymers — calculates the recoverable energy from pyrolysis and dispatches the item to the nearest suitable facility.

6.1 Hardware and Wiring

Component	Connection	Purpose
TCS34725 RGB Sensor	I2C — SDA → GPIO 21, SCL → GPIO 22	Detects object placement (Clear/Lux drop) and provides the colour signature used to classify the material.
Load Cell + HX711	DOUT → GPIO 27, SCK → GPIO 14	Measures the ingested item's mass (M_i), used for both scrap valuation and pyrolysis-yield calculations.
HC-SR04 Ultrasonic	Trig → GPIO 5, Echo → GPIO 18	Monitors storage-bin fill level to trigger the bin-full lockout.
SG90 Routing Gate Servo	PWM → GPIO 12	Swings to direct the sorted item toward the market bin, the compactor, or the pyrolysis feedstock bin.
MG996R Crusher Servo	PWM → GPIO 15	Simulates the metal compaction press on a HOLD decision.

Status LEDs (Green / Yellow / Red)	GPIO 19 / 23 / 4	Green = SELL, Yellow = HOLD / Crushing, Red = Bin Full.
Active Buzzer	GPIO 13	Audible confirmation of a scan, and a continuous alarm on bin-full lockout.

6.2 Backend Logic

- Material classification: the FastAPI backend classifies the item as metal or polymer directly from the raw R, G, B and Clear/Lux channels.
- Base metals (Aluminium, Copper, Steel, Lead, Nickel, Zinc): a 90-day ARIMA/LSTM forecast produces a SELL or HOLD decision. SELL routes to the market-liquidation bin; HOLD routes to the compactor and activates the crusher servo.
- Non-recyclable polymers (e.g. PVC): the backend computes the net energy recovery potential using $E_{rec} = M_i \times LHV_i \times \eta$ (LHV_{PVC} = 20.0 MJ/kg, η = 67%), then assigns the nearest Sri Lankan waste-to-energy or pyrolysis facility (e.g. Kerawalapitiya WTE Plant, Karadiyana pyrolysis pilot site).
- Bin-full lockout: once the HC-SR04 reading falls below the 5 cm safety threshold, the backend rejects further ingestion requests until the lockout is explicitly cleared after the bin is dispatched.

6.3 Display

Live results — forecasted price, SELL/HOLD decision, calculated energy recovery, target facility, and lockout status — are shown on a web dashboard that polls the backend, in place of the physical OLED originally specified for this component.

7. Unified System Integration

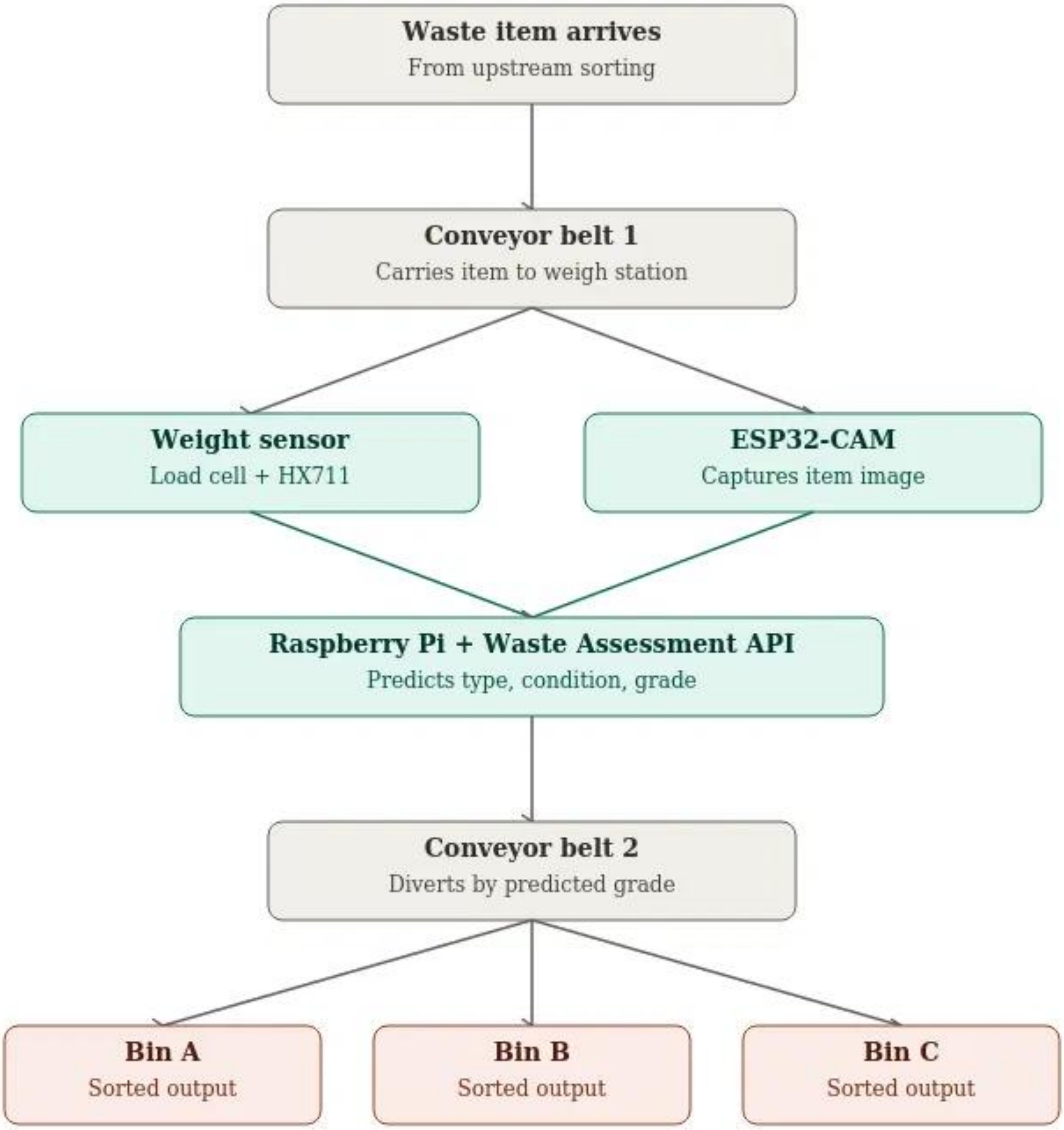
All four components converge on a shared FastAPI backend and a unified web dashboard. The table below summarises how each edge controller communicates with that shared layer.

Component	Controller	Uplink	Data Sent	Data Returned
1	ESP32 DevKit V1	Wi-Fi + MQTT	Gas/temperature/humidity readings, alert state	Acknowledgement / threshold updates
2	ESP32	Wi-Fi (JSON)	Moisture, distance, colour reading	Sorting process recipe
3	Arduino UNO → Raspberry Pi	Serial, then HTTP	Weight reading; captured image	Waste type, condition, confidence, decision
4	ESP32 DevKit	Wi-Fi (HTTP POST)	Raw R/G/B/Clear, weight, bin distance	SELL/HOLD or route decision, bin-lockout state

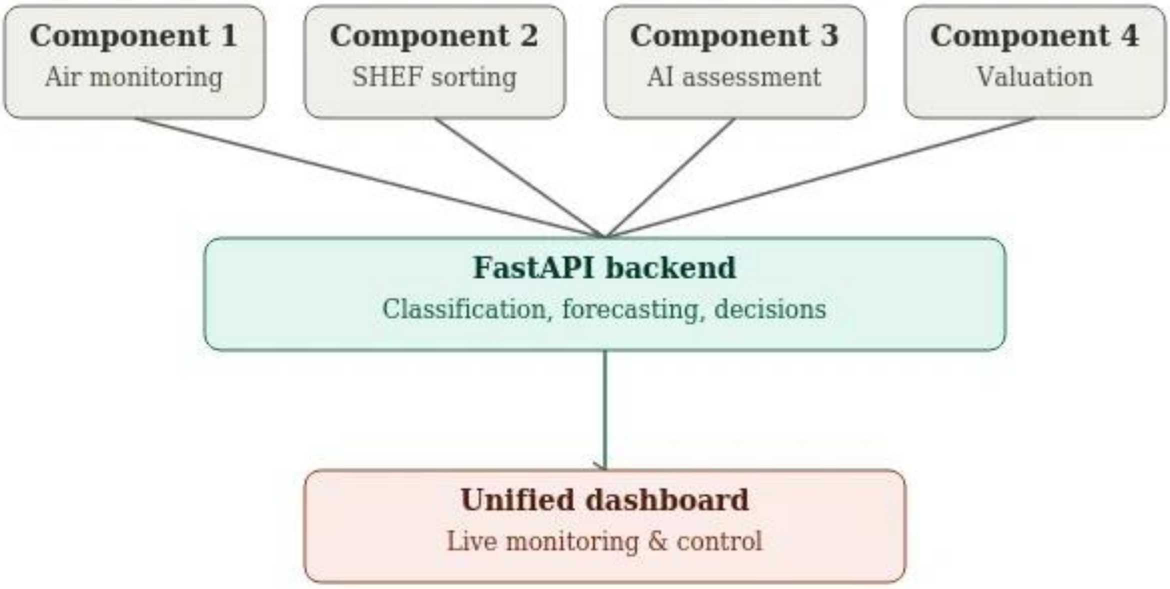
Each component's decision loop is independent — a fault or delay on one edge device does not block the others — while the shared backend gives the project a single point from which to monitor, log, and evaluate the whole system's performance during the panel demonstration.

8. Conclusion

Together, the four components form a complete sense-decide-act loop for automated waste handling: Component 1 protects the working environment, Component 2 sorts plastics by condition, Component 3 supplies AI-driven waste identification, and Component 4 turns that identification into an economic and environmental disposition decision. Each edge device is deliberately simple and inexpensive, with the analytical complexity concentrated in the shared backend — keeping the physical prototype affordable and maintainable while still demonstrating the full research contribution of each team member.



Gray = conveyor/structural, teal = sensing & AI processing, coral = output bins



Gray = component edge devices, teal = backend, coral = dashboard